

Unit 4.5 Curve Sketching. p315

Note Title

2014/04/12

Guideline for sketching a curve p315

- f { D Domain
 I Intercepts
 S Symmetry (Odd / Even)
 A Asymptotes (Horizontal / Vertical)
- f' { D Decreasing / Increasing intervals.
 E Extreme values (Max / min)
- f'' { C Concavity
 I Inflection points

Examples: Draw the curve of

① $f(x) = \frac{x^2}{x^2 - 4}$

1. $x^2 - 4 \neq 0 \Rightarrow x \neq \pm 2$
 $\Rightarrow D_f = \mathbb{R} \setminus \{\pm 2\}$

2. $x = 0 \Rightarrow y = 0$ (0,0)
 $y = 0 \Rightarrow x = 0$

3. $f(-x) = \frac{(-x)^2}{(-x)^2 - 4} = \frac{x^2}{x^2 - 4} = f(x)$
 f is even.

4. $\lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - 4} \left(\frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow -\infty} \frac{2x}{2x} = 1$
 $\lim_{x \rightarrow +\infty} \frac{x^2}{x^2 - 4} \left(\frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow +\infty} \frac{2x}{2x} = 1$

So $y=1$ is a horizontal Asympt.

$$\lim_{x \rightarrow 2^+} \frac{x^2 \oplus}{(x-2)(x+2)} = +\infty$$
$$\lim_{x \rightarrow 2^-} \frac{x^2 \oplus}{(x-2)(x+2)} = -\infty$$

} $x=2$ is V.A.

Check!!

$$\lim_{x \rightarrow -2^+} f(x) = -\infty$$
$$\lim_{x \rightarrow -2^-} f(x) = +\infty$$

} $x=-2$ is V.A.

Q10

5. $\frac{dy}{dx} = \frac{-8x}{(x^2-4)^2}$

f decreases if $f' < 0$

$$\therefore \frac{-8x}{(x^2-4)^2} < 0$$

$$\Rightarrow x > 0$$

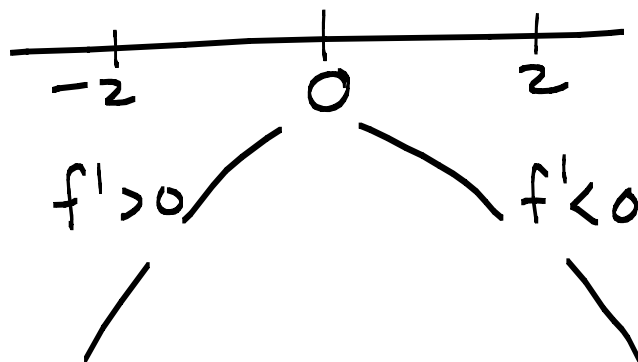
f decreases on $(0, \infty)$ and

f increases on $(-\infty, 0)$

6. loc max at $x=0$

$$f' = 0 \text{ at } x=0$$

So CV is $x=0$



$$7. \text{C} \quad f''(x) = \frac{24x^2 + 32}{(x^2 - 4)^3} \quad (\text{Check!!})$$

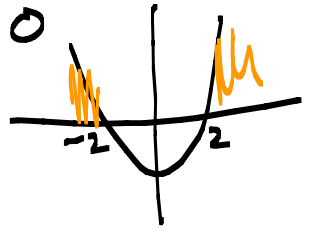
$f''(x) > 0$ the f Conc. Up.

$$\frac{24x^2 + 32}{(x^2 - 4)^3} > 0 \Rightarrow (x^2 - 4)^3 > 0$$

$$(x^2 - 4)(x^2 - 4)^2 > 0$$

$$\Rightarrow x^2 - 4 > 0$$

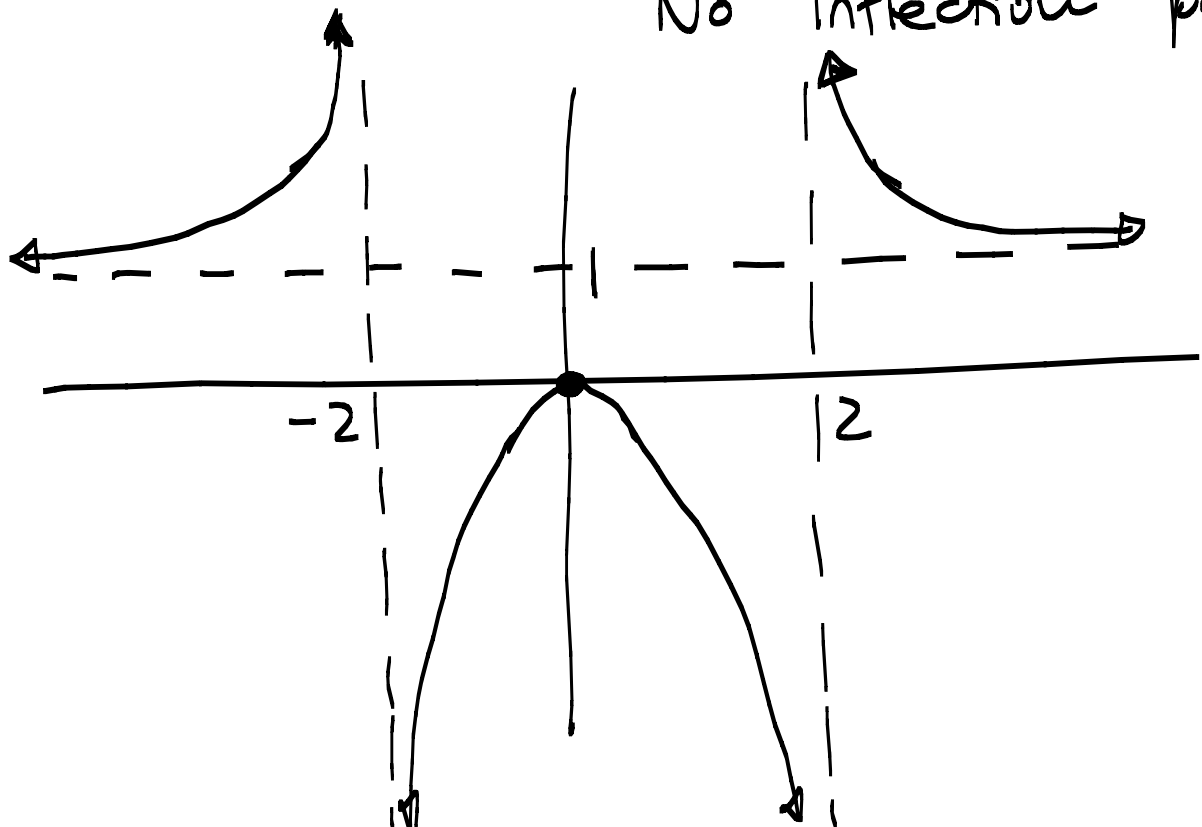
f concave up on $(-\infty, -2) \cup (2, \infty)$
 concave down on $(-2, 2)$



$$8. f''(x) = 0 \quad \frac{24x^2 + 32}{(x^2 - 4)^3} = 0$$

$$\Rightarrow 8(3x^2 + 4) = 0$$

No inflection points!



② $f(x) = \frac{e^{2x}}{x+1}$ Sketch the graph using all the information in DISADECI

D 1. Domain is $\mathbb{R} \setminus \{-1\}$ (0,1)

I 2. $f(0) = 1$, no x -intercept.

S 3. Not Symmetric

A 4. $\lim_{x \rightarrow \infty} f(x) = \infty$ $\lim_{x \rightarrow -1^+} f(x) = \infty$

$\lim_{x \rightarrow -\infty} f(x) = 0$ $\lim_{x \rightarrow -1^-} f(x) = -\infty$

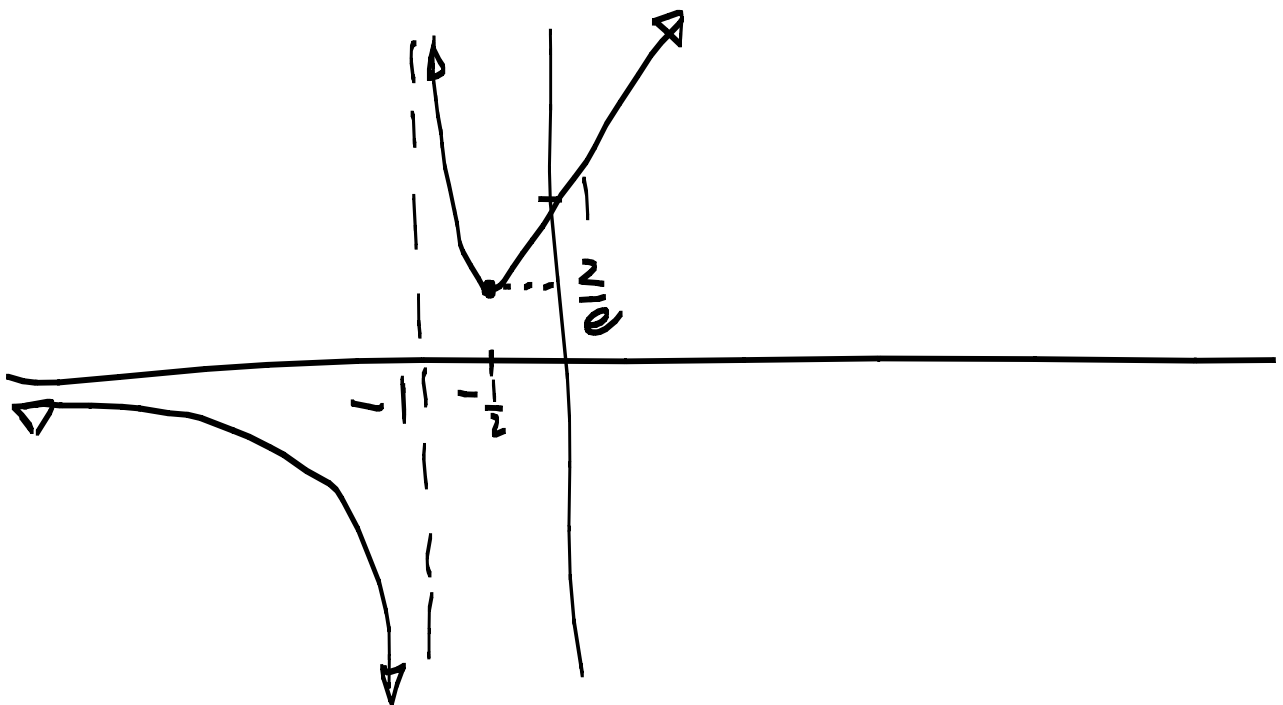
D 5. f decreases on $(-\infty, -1) \cup (-1, -\frac{1}{2})$ ||
 f increases on $(-\frac{1}{2}, \infty)$

E 6. loc min $f(-0.5) = \frac{2}{e}$

C 7. CD on $(-\infty, -1)$

CU on $(-1, \infty)$

I 8. No inflection points.



③. $f(x) = xe^{-3x}$

1. Find the extreme value(s) of f .
2. Does f have vertical asymptote(s)?
3. Does f have horizontal asymptote(s)?
4. Determine the interval(s) on which f is concave up.
5. Does f have inflection points?
6. Sketch the graph of f .

Solution:

$$1. f'(x) = e^{-3x} - 3xe^{-3x} = 0$$

$$\Rightarrow e^{-3x}(1-3x) = 0$$

$$\Rightarrow x = \frac{1}{3} \text{ since } e^{-3x} \neq 0 \text{ for all } x.$$

and f' exists on $\mathbb{R}$ so

Critical number at $x = \frac{1}{3}$

f increases if $f' > 0$

$$\Rightarrow e^{-3x}(1-3x) > 0$$

$$\Rightarrow 1 - 3x > 0 \text{ since } e^{-3x} > 0 \text{ for all } x.$$

$$\Rightarrow x < \frac{1}{3}$$

So f increases on $(-\infty, \frac{1}{3})$ and decreases on $(\frac{1}{3}, \infty)$

$\Rightarrow (\frac{1}{3}, f(\frac{1}{3})) = (\frac{1}{3}, \frac{1}{3e})$ is a local maximum.

2. No V.A. since f defined on $\mathbb{R}$.

$$3. \lim_{x \rightarrow \infty} x e^{-3x} (\infty \cdot 0) \qquad \lim_{x \rightarrow -\infty} x e^{-3x}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{e^{3x}} \left(\frac{\infty}{\infty} \right) = -\infty$$

$$\stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{1}{3e^{3x}} = 0$$

Horizontal Asympt $y = 0$.

$$4. f''(x) = -3e^{-3x}(1-3x) + e^{-3x}(-3)$$

$$= -3e^{-3x}(2-3x)$$

f concave up if $f'' > 0$ so

$$-3e^{-3x}(2-3x) > 0$$

$$\Rightarrow 2-3x < 0 \Rightarrow x > \frac{2}{3}$$

f is concave up on $(\frac{2}{3}, \infty)$ and
concave down on $(-\infty, \frac{2}{3})$

5. f'' --- + + +


$$f''(x) = 0$$

$$\Rightarrow x = \frac{2}{3}$$

Inflection point at $x = \frac{2}{3}$ since
 f changes from concave down to
concave up.

